

RelicVault AI — Project Write-up Summary

A crowdsourced digital heritage & minor-artifact museum · Next.js 14 · MongoDB · Zhipu GLM glm-4v-flash · OpenStreetMap · 3-language UI

Source: <https://github.com/raywang7266/RelicVault-AI/tree/main>

Problem statement	Use of AI
China has a long history and a vast amount of cultural heritage, yet many artifacts are not properly preserved, and many valuable ones remain little known. Amateur finds — village heirlooms, family keepsakes, flea-market objects — have no lightweight, AI-assisted home, so they stay unprotected and forgotten rather than becoming part of a shared archive.	AI is used in two concrete places in the product pipeline (see the flow below), plus as a coding partner throughout development. Details live in the Reflections strip.
Solution overview RelicVault AI is a full-stack web app. Users sign up, drop a photo of an artifact plus a map pin, and instantly get a structured record (name, dynasty, category, material, preservation, tags) from Zhipu GLM glm-4v-flash. Explore offers fuzzy search, #tag search, era / material / preservation filters, and a grid-or-map toggle. Coordinates blur to two decimals before storage, and protected sites are rounded to one decimal place.	1 · Vision provider (Zhipu GLM). a photo becomes a structured record (name, dynasty, material, preservation, tags) via a strict JSON prompt. Default model glm-4v-flash (free, China-direct, OpenAI-compatible API). 2 · Geocoder. a typed place name is resolved to coordinates with OpenStreetMap Nominatim — free and keyless. 3 · Coding agent. WorkBuddy scaffolds, refactors, and debugs the app end-to-end.



Green = Zhipu GLM glm-4v-flash (photo → structured record). Teal = OpenStreetMap Nominatim



GPS pin = OSM Leaflet picker; coordinates blurred to 2 decimals before storage.

Reflections

What worked	Challenges encountered	Future enhancements
• Started with the data model. The Mongoose schema came first; once right, every UI feature became a thin layer on top. • Treated the prompt as a product spec — JSON schema + 'say unknown' = auditable vision pipeline. • Picked a provider that ships: Zhipu GLM's glm-4v-flash is free and China-direct, so demo builds never hit a quota wall.	• CSS entrance animations — the AI agent performed poorly here. @keyframes stripped inside @layer; display:contents killing transforms. Three hard rules now live in my project memory. • Permissions gap on the artifacts collection (writes vs. reads) — solved at the application layer.	• Xiaohongshu-style social layer — public profiles, follow graph, direct messages. • Richer uploads — multiple images, short video, and pure-text entries. • Stronger auth — third-party sign-in (Google / WeChat), email & phone-number verification.

Impact & value. For amateur collectors and local history enthusiasts, the product lowers the cost of contributing to a shared cultural archive from days of research to under a minute; coordinates are blurred by default so dig-site secrecy is preserved. For researchers & educators, the structured JSON output is directly queryable, exportable, and mappable.